PULSE Instrument Cluster

Sprint 2 Backlog

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2026-09-03

Contents

Document control

Field	Value
Document ID	PULSE-PLAN-002
Title	Sprint 2 Backlog
Version	0.1
Status	Draft for the sprint 1 review; frozen after review feedback is merged
Date	2026-09-03
Author	Ahmed Abdelghany
Reviewer	Cockpit Electronics / HMI Platform Group (sponsor)
Sprint window	Week 4 to week 6, 7 September to 25 September 2026
Gate	Week 6 demo: signal layer and Flutter cluster verified on live broker data
Related	PULSE-StRS-001 section 3.1, PULSE-SyRS-001, PULSE-SAF-001, PULSE-SEC-001

Change log

Version	Date	Author	Change
0.1	2026-09-03	A. Abdelghany	First draft from the delivery plan and the week-3 status columns; review feedback section left open

1. Sprint goal

Close work package 02: the signal layer and the Flutter cluster are verified on live broker data by an automated suite that runs before every demo, and the desk-only shortcuts that have a sprint 2 requirement attached are removed. The week 6 demo shows the same cluster as week 3, now with contract tests, degraded-state display, encrypted transport and pinned upstream versions behind it.

2. Capacity

Item	Value
Engineers	2 at 40 h a week
Hours in sprint	240
Committed below	196
Reserve for review feedback and risk	44

3. Backlog

Ordered by priority. Every item names the requirement it moves and the status it should reach by the week 6 gate. Estimates are engineering hours including verification.

#	Item	Moves	To status	Est. h	Verification
1	Regression suite in CI shape: broker up, signal contract, VSS regeneration, live telemetry, HMI purity, secrets, Flutter launch. Run recorded under `docs/evidence/` before every demo	CS-11, SR-9, FR-11, FR-3	Implemented	16	Suite passes on a clean checkout
2	Signal contract test against the running broker: every path in PULSE-SAD-001 appendix A subscribes and receives a typed datapoint	FR-11, FR-13, IR-2	Implemented	12	Test in the suite
3	UI smoke test per build: cluster launches, connects, renders a known injected value, exits clean; Compose cluster on the emulator	FR-12, FR-14	Implemented	20	Test in the suite
4	Staleness detection in the Flutter cluster: freshness criterion per safety-relevant signal from PULSE-SAF-001 section 3, degraded presentation, broker loss shows all stale	SR-4, SR-5	Implemented	32	Test: kill the broker, dashes appear within the criterion
5	Same staleness handling in the Compose cluster	SR-4, SR-5	Implemented	16	Test on the emulator
6	Mutual TLS on the databroker with clearly marked development certificates; start-up prints a warning whenever `--insecure` is used; both clients and both providers connect over TLS	CS-3, IR-5	Implemented	24	Suite: plaintext client refused unless the development flag is set
7	Pin the databroker image digest and the Flutter SDK version; record both in the README	NFR-8, DD-10, DD-11	Implemented	6	Inspection

#	Item	Moves	To status	Est. h	Verification
8	Software bill of materials per component (CycloneDX) and a vulnerability check in the suite	CS-8	Implemented	14	Suite
9	Driver-monitor privacy test: no file written, no socket other than the broker	CS-6	Implemented	6	Suite
10	HMI-side range and enumeration check as defence in depth, with a test that an out-of-range publish is refused by the broker	CS-5	Implemented	10	Suite
11	Catalogue checksum in the suite, so a tampered `vss_agl.json` fails the run	CS-8, T-4	Implemented	4	Suite
12	Merge the driver-monitoring branch into `main` after review; one baseline for sprint 2	NFR-5	Done	4	Onboarding guide re-run on the merged tree
13	Software architecture description for the Flutter cluster and the providers (SWAD), one page each, as promised in PULSE-SAD-001 section 1	SR-10	Implemented	16	Review
14	Safety and security review of PULSE-SAF-001 and PULSE-SEC-001 with the assigned managers; ratings confirmed or changed	SR-1, SR-2, CS-1	Reviewed	8	Review minutes
15	Virtual CAN feeder spike: `vcan0` with kuksa-can-provider and the AGL DBC, publishing three signals, to de-risk sprint 3	FR-8	Partial	8	Demonstration

4. Out of scope this sprint

Physical CAN hardware, the Raspberry Pi 5 baseline, per-client write authorisation (CS-4), security event logging (CS-7), the fail-visible renderer (SR-6) and the Autoware bridge (FR-22) stay in sprints 3 and 4 as planned in PULSE-StRS-001 section 3.1.

5. Risks carried into the sprint

Risk	Response
TLS in the Dart and Kotlin gRPC stacks costs more than estimated	Land the broker side first; a client that cannot do TLS yet keeps the development flag and the start-up warning
Staleness handling changes the visual design	Degraded presentation is specified in PULSE-SAF-001 section 3; no new design work, only the agreed dashes, greying and banner
Safety and security managers not assigned in time	Item 14 slips to sprint 3 without blocking the gate; the documents stay marked “not reviewed”

6. Review feedback, 4 September

To be filled in during the sprint 1 review. Each comment gets a line here and, where it changes a requirement, a change-log row in the owning document.

#	Comment	Raised by	Action	Target sprint

7. Definition of done for the week 6 gate

1. Regression suite passes on a clean checkout and its report is committed under `docs/evidence/`.
2. The cluster shows degraded state within the freshness criterion when the broker is stopped, on both HMIs.
3. Broker and all four clients run over mutual TLS; a plaintext start prints the warning.
4. PULSE-SyRS-001, PULSE-SAF-001 and PULSE-SEC-001 status columns updated and PULSE-RTM-001 regenerated with zero dangling references.
5. Demo rehearsed on the strip display and timed under five minutes.